

MOHAMED HOSSAM

Backend Engineer | Django & FastAPI Specialist Beni Suef, Egypt | m.hossam1050@nub.edu.eg | [Portfolio](#) | [GitHub](#) [LinkedIn](#) | Phone: +201060805271

SUMMARY

Performance-driven Backend Developer specializing in building secure, production-ready web systems using **Python, Django, and PostgreSQL**. Expert in implementing **Service Layer Architecture** and database optimization to build maintainable, scalable applications. Currently preparing for **GSoC 2026** to contribute to open-source ecosystems with a focus on high-concurrency backend systems.

TECHNICAL SKILLS

- **Backend & Architecture:** Django, Django REST Framework, FastAPI, Service Layer Architecture, System Design, RESTful APIs, Asynchronous Programming.
- **Security & Auth:** JWT/OAuth2, Session-based Auth, RBAC, CSRF/XSS Mitigation, Signed URLs (Cloudinary), Resource Hardening.
- **Databases:** PostgreSQL (Indexing, Query Optimization), MySQL, MongoDB, Redis, Schema Design.
- **DevOps & Tools:** Docker (Multi-stage builds), Nginx, Gunicorn, Uvicorn, Linux CLI, Git, Postman, CI/CD Basics.
- **Languages:** Python (Expert), JavaScript (ES6+), C++, SQL.

TECHNICAL PROJECTS

CourseHub – Scalable Course Platform | *Django, PostgreSQL, HTMX, Cloudinary* | 2025 – Present *GitHub:* [coursehub-django](#)

- Designed and developed a production-ready course platform capable of serving **10,000+ students** with secure, session-based access control.
- Built a **layered backend architecture** (Views → Service Layer → ORM) to isolate business logic, improve maintainability, and support future scalability.
- Implemented **passwordless email authentication** using UUID-based verification tokens, attempt limits, expiration handling, and secure session persistence.
- Designed a **normalized PostgreSQL schema** with indexed lookup fields (public_id, foreign keys) to optimize read-heavy traffic and concurrent access.
- Optimized database queries through controlled service-layer access, deterministic ordering, and filtered querysets to reduce unnecessary load.
- Integrated **Cloudinary CDN** for scalable media delivery, including private video streaming with signed URLs, adaptive bitrate streaming, and automatic image optimization (WebP/AVIF).
- Improved frontend-backend efficiency using **HTMX for partial page updates**, significantly reducing payload size and avoiding SPA overhead.
- Containerized the application using **Docker**, deployed with Gunicorn behind Nginx, and prepared the system for cloud and horizontal scaling.
- Applied production security best practices including **CSRF protection**, HTTP-only sessions, secure environment variables, and restricted media access.

FastAPI OCR Microservice | *FastAPI, Tesseract, Docker, Pydantic v2* | 2026 – Present *GitHub:* [FastAPI OCR Microservice](#)

- Engineered a high-performance, stateless microservice for OCR text extraction, utilizing **non-blocking I/O** by offloading Tesseract execution to thread pools to maintain event-loop responsiveness.
- Implemented **Resource Hardening** protocols including a 10MB file size limit and a 30s hard timeout to prevent CPU saturation and ensure system stability.
- Developed a **Production Observability** layer with structured JSON logging and health/readiness probes (/health , /ready) for orchestration.
- Architected a **Containerized Stack** using Docker Compose and Nginx, ensuring horizontal scalability and environment parity.

Dental Practice Management System (DPMS) | *Django, PostgreSQL* | 2025 – Present *GitHub:* [DPMS](#)

- Developed a management system for healthcare data, implementing **Role-Based Access Control (RBAC)** and secure authentication for sensitive records.
- Designed a normalized relational schema that reduced manual data handling errors by approximately **30%**.

Chess Engine & Game | *Python, Pygame, NumPy* | 2025 – Present | *GitHub:* [Chess Engine](#)

- Built a complete chess engine featuring move validation and modular architecture for future AI integration.

EDUCATION

B.Sc. in Computer Science | Nahda University, Beni Suef *GPA: 3.23/4.0 | Expected Graduation: 2028*

CERTIFICATIONS

- **CS50x:** Introduction to Computer Science – **Harvard University**
- **Backend Web Development Program** – **ALX Professional**
- **ALX Professional Foundations**

ADDITIONAL INFORMATION

- **Languages:** Arabic (Native), English (Fluent - B2).
- **Open Source:** Preparing contributions for Google Summer of Code (GSoC 2026).